

$$1) x_1 = 1, x_2 = 3, y_1 = \sqrt{x_1} = 2, y_2 = \sqrt{x_2} = 4$$

$$P_1(x) = k(x_1) + \frac{k(x_2) - k(x_1)}{x_2 - x_1} (x - x_1)$$

$$P_1(x) = 2 + \frac{4-2}{3-1} (x-1) = 2 + 1(x-1) = 2 + x - 1$$

$$2) x_1 = 2, x_2 = 4, y_1 = \sqrt{x_1} = 3, y_2 = \sqrt{x_2} = 4$$

$$P_1(x) = 3 + \frac{4-3}{4-2} (x-2) = 5 + 2(x-2) = 5 + 2x - 4$$

$$3) x_1 = -1, x_2 = 2, y_1 = \sqrt{x_1} = 4, y_2 = \sqrt{x_2} = -2$$

$$P_1(x) = \sqrt{x_2} + \frac{\sqrt{x_1} - \sqrt{x_2}}{x_2 - x_1} (x - x_1)$$

$$P_1(x) = 4 + \frac{-2-4}{2-(-1)} (x-(-1)) = 4 + \frac{-6}{3} (x+1) = 4 - 2(x+1) = 4 - 2x - 2 = 2 - 2x$$